ModelIt!

Teacher's Guide

What's the Weather Like Today?

Watching the Sky and Tracking Weather

Quick Reference

Grade Level	Kindergarten
NGSS Standard	K-ESS2-1
Time	45–60 minutes (core lesson)
Materials	Student Activity Pack, devices with ModelIt! access
Prep	Print activity packs, test ModelIt! on student devices
Grouping	Partners or small groups (2–4)

ModelIt! — Discovery Collective

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1 NGSS Standards & CAST Alignment

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Performance Expectation: K-ESS2-1

Use and share observations of local weather conditions to describe patterns over time.

Three-Dimensional Learning

Science Practice	**Analyzing and Interpreting Data** — Students collect daily weather observations and look for patterns over time, analyzing simple data about cloud cover and sunshine.
Disciplinary Core Idea	**ESS2.D Weather and Climate** — Weather is the combination of sunlight, wind, snow or rain, and temperature in a particular region at a particular time. People measure these conditions to describe and record the weather.
Crosscutting Concept	**Patterns** — Students observe and describe patterns in daily weather data, noticing that certain types of weather occur more frequently than others.

CAST Testing Alignment

2 Scoring Rubric

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Assessment Reference

This is the same rubric students see in their Activity Pack. Use it during and after the lesson for formative and summative assessment. Circle or highlight the proficiency level for each skill.

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

Scoring Guidance

Expert (4): Student demonstrates deep understanding and can transfer knowledge to new contexts.
Proficient (3): Student meets grade-level expectations and shows solid understanding.
Developing (2): Student is making progress but needs additional scaffolding or practice.
Beginning (1): Student needs significant support; consider using Differentiation strategies (Section 7).

3 Background Content Knowledge

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For Your Confidence

This section provides the science background you need to facilitate this lesson with authority. You don’t need to teach all of this — it’s here so you can answer student questions accurately.

Weather is the state of the atmosphere at a given time and place. Kindergarteners experience weather every day and can begin to observe, describe, and track it, building foundational skills in data collection and pattern recognition.

What Is Weather?

Weather describes the current conditions of the atmosphere — temperature, cloud cover, precipitation (rain, snow, sleet, hail), and wind. Weather changes from hour to hour and day to day. It is influenced by the sun's energy heating Earth's surface unevenly, the movement of air masses, and the water cycle. At the kindergarten level, students observe basic weather conditions: Is it sunny? Cloudy? Rainy? Windy? Cold or warm?

Clouds and Sunshine

Clouds are collections of tiny water droplets or ice crystals floating in the atmosphere. When there are few or no clouds, sunlight reaches the ground directly, making it bright and warm. When clouds cover the sky, they block and scatter sunlight, making it dimmer and cooler below. Thick clouds can block most sunlight and may produce rain or snow. Tracking cloud cover over time reveals weather patterns that help young scientists understand predictability in nature.

4 The LEVER Framework

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About LEVER

LEVER is how every ModelIt! lesson works. Students move through five phases to build, test, and refine a computational model. Each lesson is self-contained — students complete all five phases in one session.

L **Locate the System**
Students identify Number of Clouds as the external component (nature controls how many clouds are in the sky) and Amount of Sunshine as the internal component (it changes based on cloud cover).

E **Establish Relationships**
Students discover that more clouds lead to less sunshine — a NEGATIVE relationship.

V **Visualize & Model**
Students build a simple two-part model connecting Number of Clouds to Amount of Sunshine with a negative arrow.

E **Evaluate Outcomes**
Students test Clear Sky vs. Cloudy Sky scenarios and observe the difference in sunshine.

R **Revise & Extend**
Students explore what happens when clouds bring rain — how does rain change what we can do outside?

Dropping ModelIt! Into Your Existing Unit

ModelIt! is a **supplement**, not a replacement for your science curriculum. Each lesson stands alone, but it becomes even more powerful when students revisit their model throughout a unit. Here are easy ways to weave it in:

At the start of a unit	Use this lesson to introduce the phenomenon. Students build a "first draft" model based on what they already know. Save it.
After a reading or lab	Have students reopen their model and update it: "Based on what you just learned, would you add a component? Change a relationship?" Takes 10–15 minutes.
As a "what-if" tool	When students ask questions during class, say: "Great question — go test that in your model!" Students run new scenarios and report back.
For peer discussion	Partners compare models: "Why does your model have this arrow and mine doesn't?" Differences spark scientific debate using evidence.
As a presentation tool	Students project their model and walk the class through it live — showing components, running scenarios, answering audience "what-if" challenges in real time.
At the end of a unit	Compare Day 1 model to final model. The growth IS the assessment — students see how their thinking evolved.

5 Slide-by-Slide Facilitation

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Slide 1: Cover

2 min

Say: *"Look outside! What is the weather like today? Is the sun out? Are there clouds? Today we are going to become WEATHER WATCHERS!"*

Watch For

Walk to the window and look at the sky together. Students describe what they see using their own words.

Slide 2: What Will We Learn?

3 min

Say: *"Today we will learn about weather! Weather is what it is like outside right now. We will learn to watch the sky and find PATTERNS!"*

Watch For

Introduce 'weather' and 'pattern.' Students say each word. Show a pattern (clap-clap-snap, clap-clap-snap) and ask: What comes next?

Slide 3: Let's Be Scientists!

3 min

Say: *"Our big question is: Why is the weather different on different days? Yesterday it was ___. Today it is ___. Why does it keep changing?"*

Watch For

Quick weather vote: students hold up sun cards, cloud cards, or rain cards for today's weather. Count and graph results.

Slide 4: What Are the Parts?

5 min

Say: *"Our model has two parts. Number of Clouds is how many clouds are in the sky — nature decides that! Amount of Sunshine is how much sun reaches us — that depends on the clouds."*

Watch For

Use sorting mats. Students place Number of Clouds on OUTSIDE and Amount of Sunshine on INSIDE. Use cotton balls (clouds) and a flashlight (sun) to show blocking.

Slide 5: How Are They Connected?

5 min

Say: *"When there are MORE clouds, do we get MORE sun or LESS sun? LESS! That is a NEGATIVE connection — when one goes up, the other goes DOWN!"*

Watch For

Demonstrate: shine a flashlight (sun) on a student. Now hold cotton balls (clouds) in front. Did the light get brighter or dimmer? Students show thumbs down for negative.

Slide 6: What Did We Find Out?

5 min

Say: *"Clouds block the sunshine! More clouds means less sun. No clouds means lots of bright sunshine. And we can watch for patterns!"*

Watch For

Chant: 'When Clouds go up, Sunshine goes DOWN!' Complete the sentence frame together. This is different — it is negative!

Slide 7: STEM Challenge

10 min

Say: *"Now we are Official Weather Watchers! Every day this week we will check the sky and put a sticker on our weather calendar. Let us start TODAY!"*

Watch For

Set up a weather wall chart with days of the week. Students choose the right weather sticker for today. Plan to check every day this week.

6 Answer Key & Expected Responses

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Component Sorting

- **EXTERNAL:** Number of Clouds
- **INTERNAL:** Amount of Sunshine

Relationships

- **Number of Clouds to Amount of Sunshine:** NEGATIVE (-) — When there are more clouds, less sunshine reaches the ground because clouds block and scatter the sunlight.

Key Discoveries

- When the sky has no clouds, we get lots of warm sunshine
- When the sky is full of clouds, they block the sun and it feels cooler and darker
- Weather changes from day to day, but we can see patterns over time

7 Collaborative Learning Protocols

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Why This Matters

ModelIt! is a digital tool, but the deepest learning happens between humans. Computational modeling is most powerful when students build understanding *together* — debating relationships, challenging predictions, and constructing explanations as a community. These protocols ensure every student's voice is part of the model. Use at least ONE per lesson.

Model Gallery Walk

When: After students build models (LEVER Phase V)
Setup: Groups post their models on screens/desks. Half the class stays to "defend," half rotates with sticky notes.
Protocol: Visitors leave 2 sticky notes: one "I notice..." and one "I wonder..." After rotation, defenders read feedback aloud. Class discusses patterns.
Why it works: Students see that the SAME phenomenon can be modeled differently — and that's how real science works. (Bandura: learning through observation of peers)

Scientist Fishbowl

When: During simulation discussion (LEVER Phase E)
Setup: 4-5 students sit in center circle. Rest of class observes with clipboards.
Protocol: Inner circle discusses: "What happened in Scenario 3 and WHY?" using only evidence from their models. Outer circle records strongest arguments. Swap groups.
Why it works: Students practice scientific discourse without teacher directing. Observers learn to evaluate arguments. (Freire: dialogue as the foundation of critical consciousness)

Component Jigsaw

When: During component analysis (LEVER Phase L)
Setup: Assign each group ONE component to become "experts" on. They research it, discuss it, draw it.
Protocol: Regroup so each new team has one expert per component. Experts teach their component to the team. Team builds the model together using everyone's expertise.
Why it works: Every student is essential — the model literally cannot be built without each person's knowledge. Interdependence drives engagement. (Lave & Wenger: learning through legitimate peripheral participation)

Prediction Debate

When: Before running simulations (LEVER Phase E)
Setup: Post a scenario. Students physically move to corners: "Fire Spread will INCREASE a lot" / "INCREASE a little" / "STAY the same" / "DECREASE."
Protocol: Each corner group has 2 minutes to build their argument using model logic. One spokesperson presents. Then RUN the simulation. Losing corners explain what they missed.
Why it works: Movement + social risk + immediate feedback. Students CARE about the outcome because they committed publicly. Wrong predictions become the best learning moments.

Pair Programming for Models (Digital Collaboration)

When: Throughout model building
Setup: Two students, one device. **Driver** controls the mouse/keyboard. **Navigator** directs strategy and catches errors. Switch every 5 minutes.
Protocol: Navigator must verbalize every decision: "I think we should add a negative arrow from Rainfall to Dry Vegetation because..." Driver cannot act without Navigator's reasoning.
Why it works: Forces articulation of scientific thinking. The model becomes a shared artifact, not an individual assignment. Students who struggle with writing can demonstrate understanding verbally. Borrowed from computer science — where the best code is written in pairs.

Quick Protocol Selection Guide

Short on time (5 min)	Prediction Debate — gets bodies moving, generates energy fast
Full period available	Component Jigsaw — deep collaboration, every student contributes

After simulation day	Model Gallery Walk — peer feedback, multiple perspectives
Advanced discussion	Scientist Fishbowl — builds academic discourse skills
Every day default	Pair Programming — transforms individual work into dialogue

8 Differentiation & Extensions

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Below Grade Level

- Pre-sort components as a whole class
- Provide sentence frames for relationships
- Use visual vocabulary cards
- Pair with a strong reader

Advanced Learners

- Add 2-3 additional components
- Research real-world data
- Create multi-step causal chains
- Compare to similar systems globally

English Language Learners

- Pre-teach key vocabulary
- Provide bilingual glossary
- Use visual supports with text
- Allow verbal responses

Gifted & Talented

- Create feedback loop diagrams
- Research prevention strategies
- Write a policy brief
- Connect to climate change

9 Common Misconceptions

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Why This Matters

Addressing misconceptions proactively helps students build accurate scientific understanding. Use these as formative assessment checkpoints during the lesson.

STUDENTS MAY THINK	SCIENTIFIC UNDERSTANDING
"Clouds are made of cotton or smoke"	Clouds are actually made of tiny, tiny drops of water or bits of ice floating in the air. They form when water evaporates from the ground and rises up into the sky where the air is cool enough to turn the water vapor back into tiny droplets.

STUDENTS MAY THINK	SCIENTIFIC UNDERSTANDING
"Rain falls from holes in the clouds"	Rain happens when water droplets inside clouds bump into each other and join together, getting bigger and bigger until they are too heavy to float. Then gravity pulls them down as rain. There are no holes in clouds.